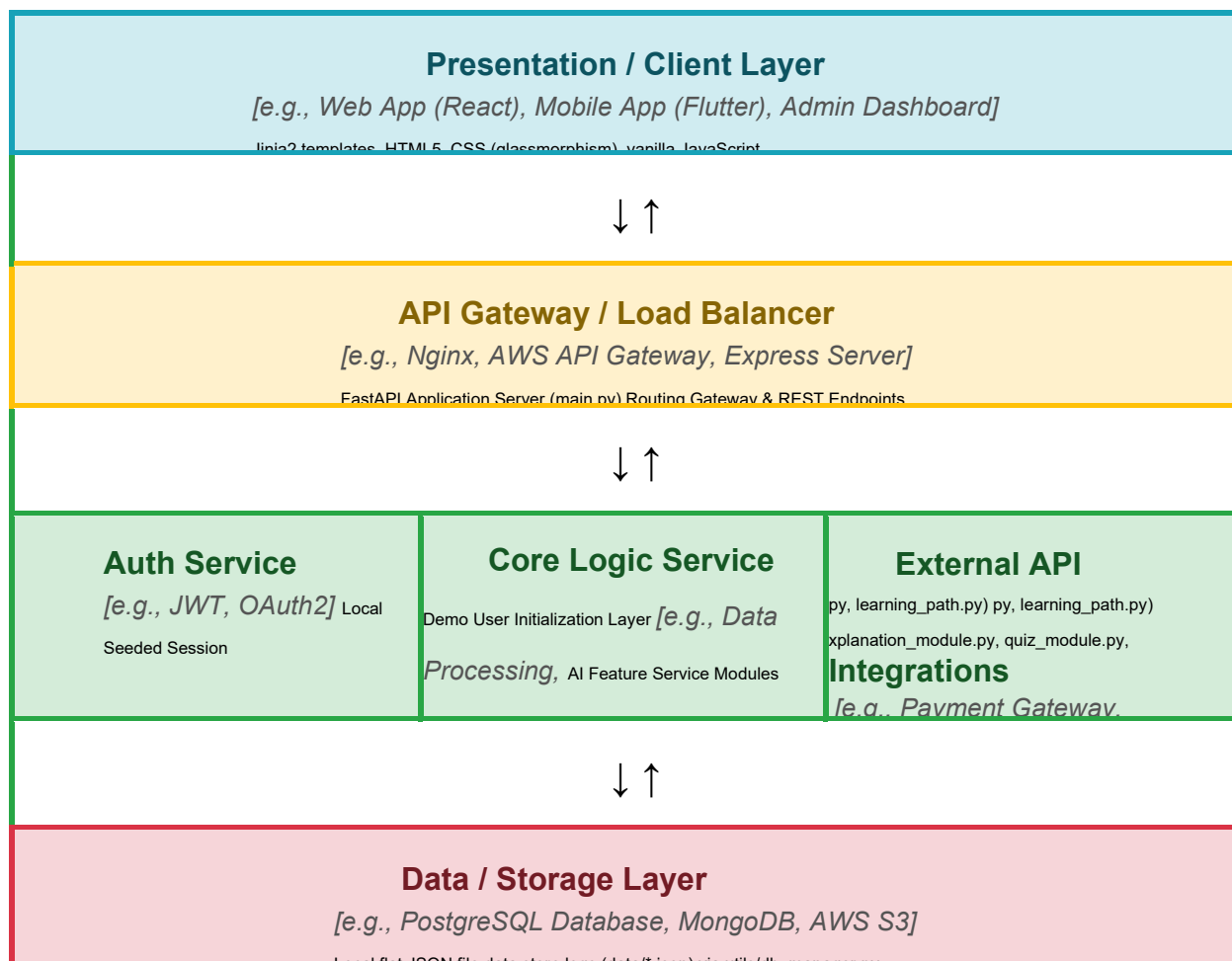


## Solution Architecture

|                      |                                                            |
|----------------------|------------------------------------------------------------|
| <b>Date</b>          | 15 March 2026 28 June 2026                                 |
| <b>Team ID</b>       | XXXXXX EduGenie - Google Gemini Powered Learning Assistant |
| <b>Project Name</b>  | XXXXXX EduGenie - Google Gemini Powered Learning Assistant |
| <b>Maximum Marks</b> | 5 Marks                                                    |

### Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



## Component Description Table

| Component Name              | Description / Role in Architecture                                                                                                   | Technologies Used                                                    |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <i>[Presentation Layer]</i> | Renders the home, chat, explain, quiz, summary, recommend,<br><i>[Describe what this component does]</i> about and                   | Jinja2 templates, HTML5, CSS,<br>vanilla <i>[List specific tech]</i> |
| <i>[API Gateway]</i>        | FastAPI app (main.py) exposing router-based endpoints (/api/ask,<br><i>[Describe what this component does]</i> /api/explain,         | FastAPI, Uvicorn, CORS middleware<br><i>[List specific tech]</i>     |
| <i>[Microservices]</i>      | Feature-specific modules (qna.py, explanation_module.py, etc.)<br><i>[Describe what this component does]</i> that build              | Python, Pydantic schemas<br>(models/sc <i>[List specific</i>         |
| <i>[Database]</i>           | utils/gemini_service.py dynamically resolves best model; <i>[Describe what this component does]</i> utils/local_t5_service.py lazily | <i>[List specific tech]</i>                                          |